INDUSTRIAL AI / INFINEON · HACKATHON 2026

Process Logic Transformer

Learning the rules of semiconductor process routes — and measuring whether they hold.

Team 123123

Volodymyr Sennikov

Ronald Juhasz

Stefan Ilic

Learn the logic, not the n-grams.

Semiconductor process routes are long, ordered sequences of **107–151 steps** across three families: **MOSFET, IGBT, IC**.

The central question we kept honest about:

Does the model learn the rules that generate valid sequences — or just memorize n-grams?

Task 1	Task 2	Task 3	Bonus
Next-step prediction	**Sequence completion**	**Anomaly detection**	**OOD 4th-family generalization**

Small transformer, big evaluation.

1 Decoder-only, from scratch

Word-level vocab (~200 tokens). No pretrained LLM — its subword machinery would hurt, not help.

2 Validator in the eval loop

Every epoch, complete held-out prefixes and score with the rule validator. A process-logic signal independent of loss.

3 Trigram baseline as the floor

Pure frequency counting tells us what 'good' means — and its OOD collapse proves local stats don't transfer.

4 Likelihood-based anomaly head

Threshold tuned on labeled data — never on the eval set.

Number normalization: a free win.

Before

```
ALIGN_MASK_LEVEL_1, ALIGN_MASK_LEVEL_2, … LEVEL_6  →
treated as 6 unrelated tokens
```

After

```
ALIGN_MASK_LEVEL  +  <N1> … <N6>  →  shared base +
number token
```

Measured effect (held-out)

Token accuracy	0.808 → 0.828	+0.020
Validation loss	0.332 → 0.300	−0.032
Model size	— → —	unchanged

A tokenizer change — not a bigger model.

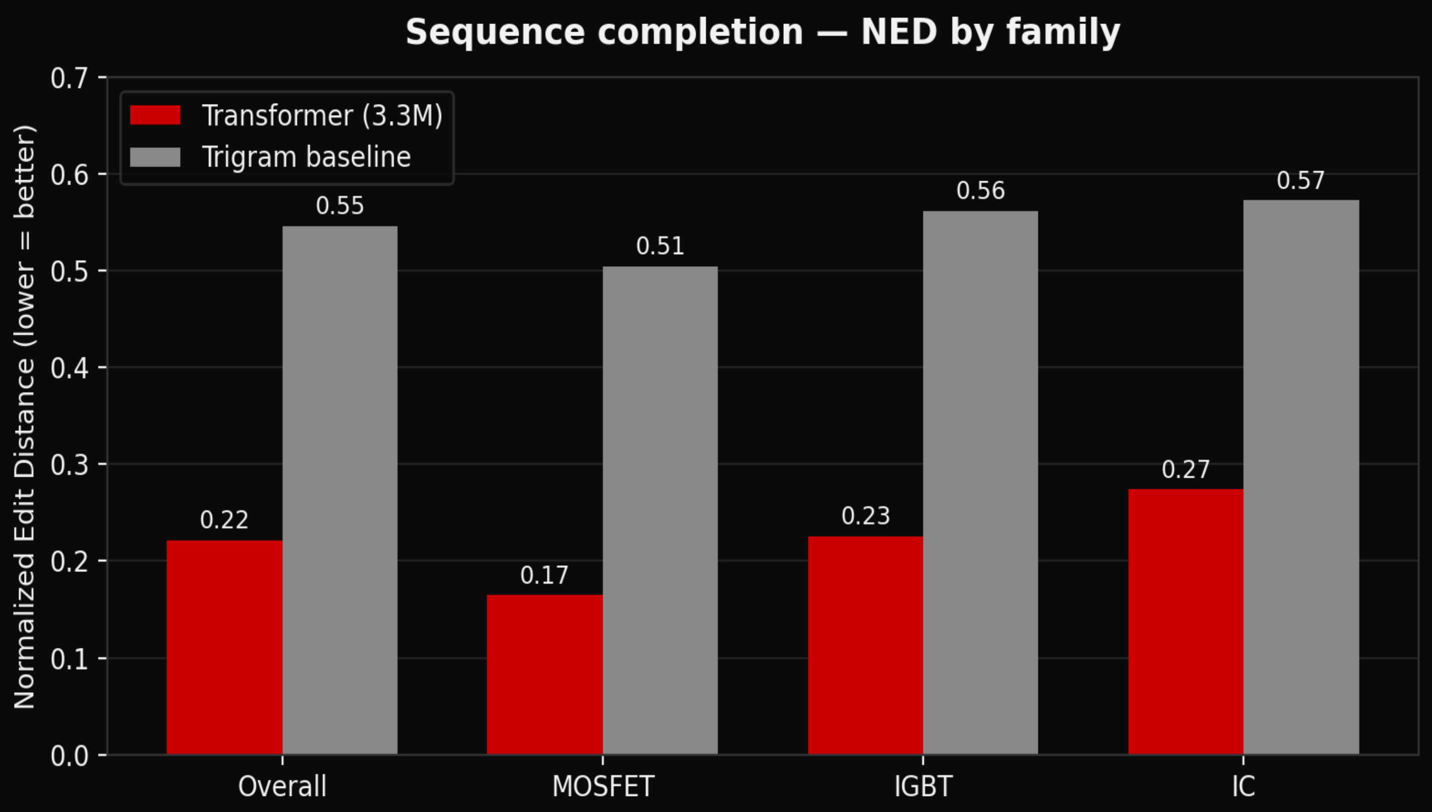
04 · PIPELINE

From synthetic data to scored submission.

Generate	Vocab	Baseline	Train	Predict	Validate
5k seqs × MOSFET / IGBT / IC	~200 tokens + number split	Trigram (CPU floor)	Decoder transformer Leonardo · 1× A100	Next-step · Completion · Anomaly (likelihood)	Rule validator + eval_metrics.py

Validator-in-the-loop: every epoch we score how often the model's completions introduce no new rule violation.

Cuts edit distance by >60% vs baseline



NED

0.22

vs 0.55 baseline

Token acc.

0.42

vs 0.32 baseline

Per-family wins

3 / 3

MOSFET · IGBT · IC

450 held-out examples · new seeds · same metrics as official eval_metrics.py

Strong on anomaly. Saturated on next-step.

ANOMALY DETECTION · TASK 3

F1 = 0.96

Model-likelihood scoring with a threshold tuned on labeled data — never on the eval set.

Applied to 987 official sequences → submission/anomaly.csv

NEXT-STEP · TASK 1

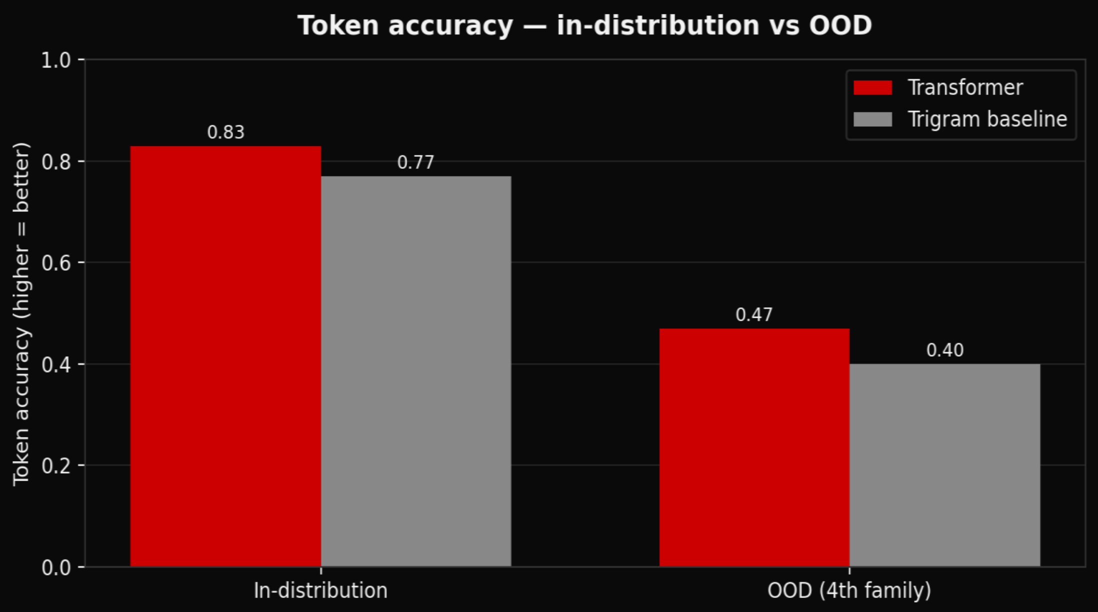
Metric	Transformer	Trigram
Token accuracy	0.83	0.77
Top-5 accuracy	0.99	0.99

Top-5 accuracy saturates near ceiling and is therefore weak as a differentiating metric.

What worked. What didn't.

WHAT WORKED

- Completion: NED 0.22 vs 0.55 — wins on all 3 families.
- Anomaly: F1 = 0.96 on labeled data.
- Tokenizer trick: free +0.02 accuracy, −0.03 loss.
- Validator-in-loop gave a process-logic signal beyond loss.
- Baseline kept us honest about saturated metrics.



WHAT DIDN'T

Scaling 3.3M → 25.5M didn't move accuracy (~0.83 ceiling). OOD weak: 0.47 vs 0.40 baseline — honest negative.

What we'd do with more time.

Randomize generation marginals

Vary optional-step probabilities and cycle counts so the model isn't calibrated to one distribution — likely the biggest OOD lever.

Discriminative anomaly head

Train directly on labeled valid/invalid examples instead of relying on generative likelihood.

Full scaling sweep

Run tiny / baseline / large / xl as four parallel single-GPU jobs and report a proper scaling curve.

THANK YOU

Questions?

LIVE DEMO

```
streamlit run streamlit_process_dashboard.py
```

Side-by-side: trigram baseline vs. trained transformer on identical inputs.

CREDITS

Libraries: PyTorch · NumPy · pandas · Plotly · Streamlit

Infrastructure: Leonardo (EuroHPC / CINECA) · SLURM · pixi

Pretrained models: none — trained from scratch. **External APIs:** none.

Team 123123 · Volodymyr Sennikov · Ronald Juhasz · Stefan Ilic